

Statistical Models in Computational Biology

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Consider a single-cell transcriptomics data set, where cells of different types have been sequenced. We expect that the cell type can be well distinguished by looking at the abundance of RNA molecules of a particular kind.

Lets assume we have this specific molecule count $x_i \in \{0, 1, 2, \dots\}$ for each cell in the data set. Given the cell type, we treat the counts x_i as i.i.d observations from a negative binomial distribution. Further, we do know that in our sample there are at most K distinct cell types.

Your task will be to estimate the component parameters p_k of different cell types, as well as their corresponding proportions π in the data set using variational inference (Mean field - Coordinate ascent variational inference (CAVI)).

For this you will derive the variational inference updates for a mixture of negative binomial distributions. Second you will implement your derived updates to estimate the parameters on a toy example in R.

Problem 17: Derivation of variational inference updating equations for a negative binomial mixture of cell populations (5 points)

We assume a dataset of counts $\{x_i\}_{i=1}^N$ generated from a mixture model with $K = 2$ negative binomial components. Each observation is associated with a latent indicator variable z_i denoting its generating component.

Consider the following model:

Likelihood:

For each observation x_i , we assume

$$x_i \mid z_i = k, p_k \sim \text{NegBin}(r, p_k), \quad P(x_i \mid z_i = k, p_k) = \binom{x_i + r - 1}{x_i} p_k^r (1 - p_k)^{x_i},$$

where $r > 0$ is a fixed and known dispersion parameter.

Latent Variables and Priors:

The latent indicators $z_i \in \{1, 2\}$ is drawn from a categorical distribution with mixing proportions $\pi = (\pi_1, \dots, \pi_K)$:

$$z_i \mid \pi \sim \text{Categorical}(\pi), \quad P(z_i = k \mid \pi) = \pi_k.$$

A Dirichlet prior is placed on π :

$$\pi \sim \text{Dirichlet}(\alpha_0), \quad P(\pi) = \frac{1}{B(\alpha_0)} \prod_{k=1}^K \pi_k^{\alpha_{0,k}-1}.$$

Each negative binomial parameter p_k is given a Beta prior:

$$p_k \sim \text{Beta}(a_0, b_0), \quad P(p_k) = \frac{1}{B(a_0, b_0)} p_k^{a_0-1} (1-p_k)^{b_0-1}.$$

(a) Write down the full joint probability & log-probability of the data and latent variables.
(2 points)

Next we want to approximate the posterior with a variational distribution that factorizes as:

$$Q(\pi, p, \mathbf{z}) = Q(\pi) \prod_{k=1}^K Q(p_k) \prod_{i=1}^N Q(z_i).$$

We choose:

- $Q(\pi)$ as Dirichlet,
- $Q(p_k)$ as Beta, and
- $Q(z_i)$ as Categorical.

The evidence lower bound (ELBO) is:

$$\mathcal{L}(q) = \mathbb{E}_Q[\ln P(\mathbf{x}, \mathbf{z}, \pi, p)] - \mathbb{E}_Q[\ln Q(\mathbf{z}, \pi, p)].$$

Using coordinate ascent variational inference, the update for each factor is:

$$\ln Q^*(\theta_j) \propto \mathbb{E}_{Q(\theta_{-j})}[\ln P(\mathbf{x}, \theta)],$$

where θ_j denotes one group of variables.

(b) Derive the variational update for $Q(z_i)$ (1 points)

(c) Derive the variational update for $Q(\pi)$ (1 points)

(d) Derive the variational update for $Q(p_k)$ (1 points)

For b, c, and d you should obtain solutions proportional to:

- $Q(z_i)$:

$$Q(z_i = k) \propto \exp \left\{ \psi(\alpha_k) - \psi \left(\sum_{j=1}^K \alpha_j \right) + r \left[\psi(a_k) - \psi(a_k + b_k) \right] + x_i \left[\psi(b_k) - \psi(a_k + b_k) \right] \right\},$$

with $\sum_{k=1}^K Q(z_i = k) = 1$. Please note that ψ is the digamma function (R-base: `digamma(x)`).

- $Q(\pi)$

$$Q(\pi) = \text{Dirichlet}(\alpha_1, \dots, \alpha_K), \quad \alpha_k = \alpha_{0,k} + \sum_{i=1}^N Q(z_i = k).$$

- $Q(p_k)$:

$$Q(p_k) = \text{Beta}(a_k, b_k), \quad a_k = a_0 + r \sum_{i=1}^N Q(z_i = k), \quad b_k = b_0 + \sum_{i=1}^N Q(z_i = k)x_i.$$

With the updating equations at hand we can iterate through them until the ELBO converges.

Problem 18: R-Implementation of variational inference updating equations for a Negative Binomial mixture of cell populations (5 points)

You are given data on single-cell analyses of metastatic bone marrow in human neuroblastoma [1]. Neuroblastoma is an aggressive pediatric cancer with a high rate of metastasis to the bone marrow. Understanding the cellular and molecular mechanisms of the metastatic tumor microenvironment is crucial for developing new therapies. Single-cell RNA-Seq has been performed to characterize immune and tumor cell alterations in neuroblastoma bone marrow metastases. Your data includes an excerpt of only B-cells and T-cells with the expression level of *ATP6V0E1* for every cell. Your task is now to implement the previously derived Negative binomial mixture model to derive the cell annotation. Please use the provided solutions from before (1: b, c, d).

(a) Implement the above derived variational inference procedure to estimate λ_k and π for the data in `vi.csv`. (5 points)

- Load `vi.csv` into R.
- Evaluate the full VI procedure for dispersion parameters $r \in \{10, 100, 1000\}$
- Define vague priors ($a_0 = 1, b_0 = 1, \alpha_0 = \mathbf{1}$)
- Randomly initialize $Q(z_i = k)$ (uniform)
- Iterate through the derived updating equations (≈ 1000 iteration)
- Report p_k and the relative fraction of cells π

References

- [1] Shenglin Mei, Adele M Alchahin, Bethel Tesfai Embaie, Ioana Maria Gavriliuc, Bronte Manouk Verhoeven, Ting Zhao, Xiangyun Li, Nathan Elias Jeffries, Adena Pepich, Hirak Sarkar, et al. Single-cell analyses of metastatic bone marrow in human neuroblastoma reveals microenvironmental remodeling and metastatic signature. *JCI insight*, 9(6):e173337, 2024.